TRAN AN PHU

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EDUCATION

Hung Vuong Junior High School 2020 - 2024
Le Thanh Tong Primary, Junior and Senior High School 2024 - 2027

GPA: Grade 9: **9.0** Grade 10: **9.0** Grade 11: **9.0**
Scores: SAT: **1520** IELTS: **7.5**
AP Statistics: **4** AP Calculus BC: **5** AP Calculus AB: **4** AP Computer Science A: **4**

HONORS AND AWARDS

- **Grand Prix | AI-JAM US - 11th International AI Invention Challenge | 2026**
  (ROV for Aquatic Integrity - selected among 840 teams from 41 countries)
- **Gold Award & ESG LOHAS Special Award | International Creative Papers Conference & Olympic, South Korea | 2026**
  (Landslide Early Warning System Using Machine Learning)
- **Gold Medal | 1 IDEA 1 WORLD International Innovation Competition, Istanbul | 2026**
  (Machine Learning-Based Safety Compliance Classification Using Camera Vision)
- **Gold Medal | Silicon Valley International Invention Festival | 2025**
  (AI-Integrated Nutritional Care Software with Online Specialists - 200+ inventions, 25 countries, five continents)
- **Gold Award | Southeast Asia Computer Science Olympiad (SEACSO) - National Round | 2025**
  (Ranked in the top tier for advanced syntax coding among high school students in the regional computer science circuit)
- **Bronze Award** | Singapore Math Global Finals | 2025
  (3,686 contestants from 1,373 schools worldwide)
- **Exemplary Award | Singapore Math Challenge | 2024**
  (Top 8%, Grade 9)

PASSION PROJECT

AI-Based Wildlife Monitoring Drone
Student Researcher and Prototype Developer *2026*

- Inspired by horseback riding and his observations of changes in horses' gait, explored how aerial footage and AI could help monitor animal movement.
- Designed and built a camera-equipped quadcopter prototype with onboard computing, flight-control, and GPS components.
- Collected video and field data on horses at a riding school and on wildlife, including elephants and gaur, in Mã Đà Forest.
- Annotated wildlife images in Roboflow, trained a YOLOv26 object detector, and developed a workflow to pair image detections with UAV telemetry for ground-station review.

Machine Learning-Based Safety Compliance Classification Using Camera Vision
Model Development and Training

- Developed and trained the model that classifies camera images as Helmet or No Helmet to support workplace safety monitoring.
- The project received a Gold Medal at 1 Idea 1 World in Istanbul in May 2026.

AI-Integrated Nutritional Care Software with Online Specialists April–August 2025
Research Team Member, Database Management

- Helped develop an AI-supported personalized nutrition platform with more than 350 EatClean recipes, nutrition data for more than 700 foods, and an online specialist chat feature.
- Applied data analysis and predictive modeling to support personalized recommendations; managed a MySQL database to organize nutrition data and improve data processing.
- The project received a Gold Medal at the 2025 Silicon Valley International Invention Festival in the United States.

RESEARCH

A Multi-View Anomaly Detection Framework for Elephant Movement Based on GPS Data Jan 2026 - June 2026
Co-author | The Indonesian Journal of Computer Science | DOI: 10.33022/ijcs.v15i3.5146

- Preprocessed 12,170 elephant GPS records and engineered 25 spatial, temporal, and movement features for anomaly analysis.
- Trained a compact Random Forest model using nine selected features, SMOTE, undersampling, and class weighting to address class imbalance.
- Contributed to research presenting a lightweight Random Forest framework that reported 97% test accuracy and 0.99 ROC-AUC, with 9 selected features and 16-bit data quantization to support resource-efficient wildlife monitoring.

TECH-DRIVEN PRODUCTS

EduMath Lab | Educational Technology Platform Developer December 2025–Present
Tech Stack: React Native, TypeScript, Expo, Next.js, REST APIs | GitHub

- Developed an interactive mathematics learning platform for children ages 5–12, integrating a mobile application with a web-based administration system for managing educational content.

- Programmed 5 math mini-games covering arithmetic, memory matching, numerical patterns, and real-world problem-solving; integrated quizzes, automated feedback, progress tracking, and leaderboards.
- Introduced the platform at Hi Vong School in Ho Chi Minh City, providing 30 learning accounts to 30 students; proposed a US$10/month subscription model to donate future revenue to local shelters.

Memory Challenge | Mobile App Developer October 2024–September 2025
Tech Stack: React Native, TypeScript, Expo, Redux Toolkit | GitHub
- Developed a brain-training mobile application featuring 2 interactive games and 60 progressive levels designed to challenge memory, visual recognition, and observation skills.
- Programmed randomized card-matching and color-detection algorithms with adjustable difficulty, countdown timers, star-based scoring, achievement badges, and progress tracking.
- Conducted user testing with 50 elementary and middle school students, collected feedback on gameplay and usability, and refined the application based on user experiences.

LEADERSHIP AND COMMUNITY ENGAGEMENT

Tekite Computer Science Club | Fanage August 2024–Present
President, 20-Member Club
- Coordinated members, technology projects, and STEM outreach; initiated and ran a science fair for 50 primary school students in Núi Tượng, Đồng Nai, featuring water rockets, light refraction, and water-pump experiments.

SENTEM Đồng Tháp 2025 Summer 2025
Project Lead and Volunteer Instructor
- Designed and taught workshops on water-rocket assembly and programming Arduino-based Bluetooth cars; organized rocket launches and car races for nearly 100 students.
- Helped raise and distribute 100 gift packages and 20 scholarships; the program reported a total value of VND 30 million.

Summer Computing and Engineering Outreach Summer 2026
Tekite Organizing Team Member
- Taught Scratch game programming to 30 students facing financial hardship at Hi Vong School in District 6, Ho Chi Minh City.
- Worked with three craft and science clubs on a camp for 200 ethnic-minority primary students at Vo Thi Sau Primary School in Dak Lak Province; Tekite led model race-car assembly and a remote-controlled obstacle-course race.
- The joint program distributed 200 learning-supply packages and served a buffet meal, with a reported total value of VND 50 million.

Green Defender | Fanpage July 2025–Present
President
- Partnered with Cat Tien National Park on a conservation learning visit for 20 members covering forest ecosystems, wildlife, hydroponics, and supervised animal-care activities.
- Worked with the Đồng Nai Nature and Culture Reserve Management Board on a habitat-restoration program that planted 300 trees; the reported project value was VND 30 million.

TECHNOLOGY TRAINING

Ho Chi Minh City University of Technology July–August 2024
IoT Program Participant
- Studied IoT and helped develop a radar-sensor obstacle-avoiding smart car and a smart-home project.

Saigon Hi-Tech Park June–July 2025
Technology Program Participant
- Practiced robotics, AR/VR, AI, and 3D scanning; studied industrial production processes and assessed potential market applications and product ideas.
- Developed and presented an idea to use AI to inspect drill-bit quality for ANMI.

CULTURAL PROJECT

Flow of the Folk August 2024
Water Puppet Performer and Organizer
- Studied Vietnamese water puppetry and performed at a fundraising event for 50 attendees at Phương Nam Water Puppet Theater; the event raised VND 20 million for Tâm Đức Shelter.
- Helped introduce water puppetry to children at Tâm Đức Shelter through a cultural exchange.